

Computational Mechanics:

1. Overview & Scope

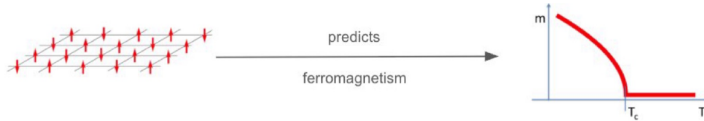
ILIAD INTENSIVE — SPRING 2026

Outline:

1. What is Computational Mechanics?
2. Computational Mechanics & AI Safety
3. Overview — where is the program at?
4. Scope — what will we tackle today?

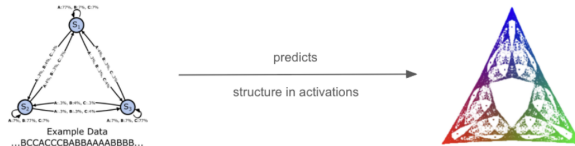
What is Computational Mechanics?

- * Statistical Mechanics makes predictions about the behaviour of systems by considering their statistical properties. E.g.



Figures from
Yuvan et al.

- * Computational Mechanics (comp-mech) makes predictions about the behaviour of systems by considering their computational properties. E.g.



Figures from
Shai et al.

What is Computational Mechanics?

Some questions that are central to comp-mech:

- * How much of the past does a system store?
- * How does the system store this information?
- * How does this information affect system behaviour?

We will consider each of these questions for transformers today!

What is Computational Mechanics?

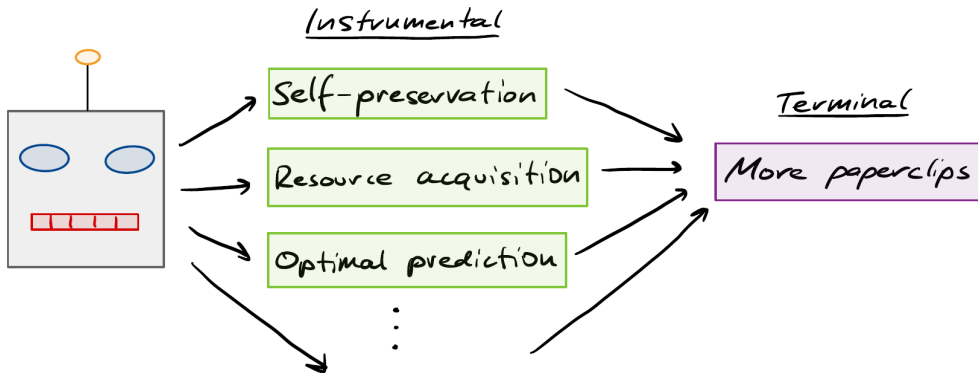
Not all systems are interesting! Comp-mech places an emphasis on those performing optimal prediction. In this setting the central question becomes:

What are convergent structures relevant for
optimal prediction?

Insight on this question greatly constrains our search space when considering optimal predictors.

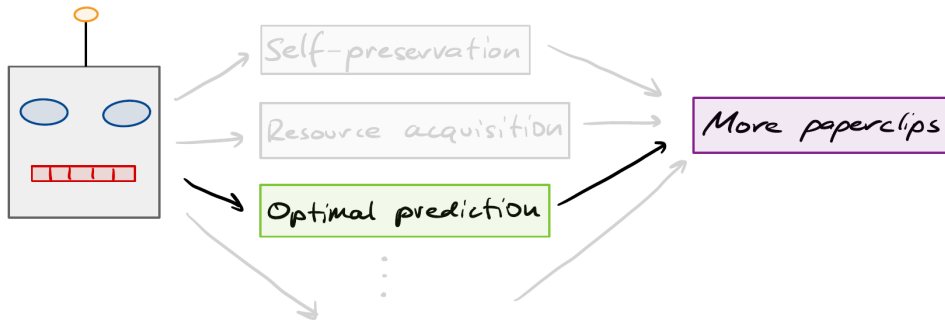
Computational Mechanics & AI Safety

We have strong reason to believe that AI systems will pursue a range of instrumental goals in service of their terminal goal



Computational Mechanics & AI Safety

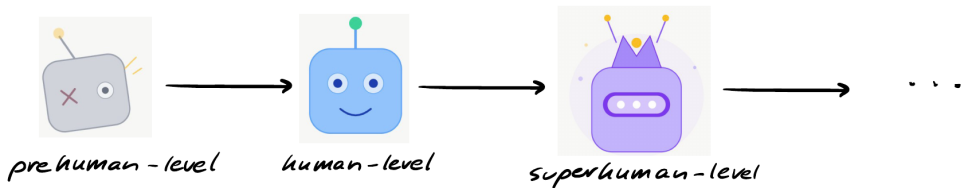
Comp-mech uses an instrumental goal as an intervention target point



By considering internal representations consistent with optimal prediction the search space is reduced.

Computational Mechanics & AI Safety

Targeting an instrumental goal is scale-free — it will be relevant to current and future models



Which contrasts scale-sensitive approaches e.g. AI control that are only applicable up to a fixed capability scale.

Moreover, as capabilities increase models will better approximate optimal predictors.

Overview — where is the program at?

Comp-mech applied to AI safety is a young research program (~ 2 years). There is much to do!

Progress can be broken down into two orthogonal directions:

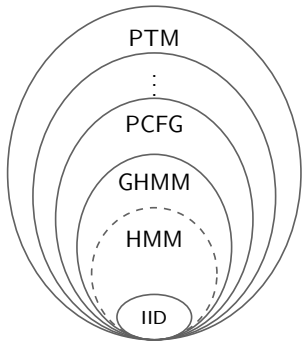
- * For what classes of data generating processes do we understand optimal prediction?
- * For what scale of neural networks have we identified the structures of optimal prediction?

Processes:

Fix a (computable) stochastic process and consider:

What resources are required to generate this process?

This naturally leads to a hierarchy:



PTM Probabilistic Turing Machine

PCFG Probabilistic Context-Free Grammar

GHMM Generalised Hidden Markov Model

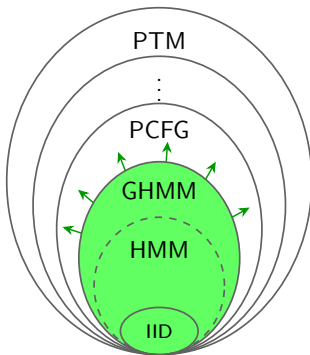
HMM Hidden Markov Model

IID Independent & Identically Distributed

(non-exhaustive)

Processes:

For what classes of data generating processes do we understand optimal prediction?



PTM Probabilistic Turing Machine

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Neural Networks:

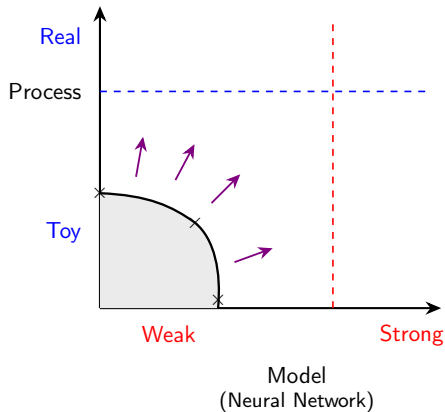
For what scale of neural networks have we identified the structures of optimal prediction?

- * Transformer: $\sim 10^5$ parameters
- * LSTM: $\sim 10^5$ parameters
- * RNN: $\sim 10^4$ parameters
- * GRU: $\sim 10^5$ parameters

Very toy compared to current frontier models ($\sim 10^{12}$)!

Overview — where is the program at?

We are working to push the Pareto frontier:



Overview — where is the program at?

We are working to push the Pareto frontier:

Real ↑

Wild Speculation:

As the research program is scale-free one view is that we should develop the foundations such that future automated AI Safety researchers can pursue these ideas independently

Weak

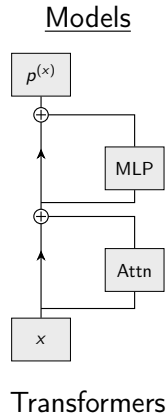
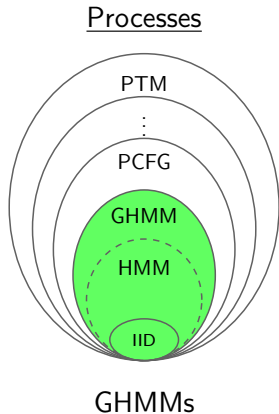
Strong

Model
(Neural Network)



Scope — what will we tackle today?

We will take a slightly narrower view and consider:



Plan:

- | | |
|---|---------------------|
| 1. Foundations — Hidden Markov Models | [lecture/exercises] |
| 2. Foundations — Prediction | [lecture/exercises] |
| 3. Applications — Designing Processes | [tutorial] |
| 4. Applications — Belief geometry in transformers | [reading] |
| 5. Applications — How do transformers do it? | [tutorial] |
| 6. Q & A with Adam Shai and/or Paul Riechers | |

Questions?

Computational Mechanics:

2. Hidden Markov Models

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Outline:

1. What is a hidden Markov Model (HMM)?
2. HMM minimality & uniqueness
3. Generalised hidden Markov Model (GHMM)

What is a hidden Markov Model?

Intuitively, a hidden Markov Model (HMM) is a stochastic process that consists of two components:

- * A 'hidden' unobserved component that evolves according to a Markovian dynamic — the next hidden state only depends on the current hidden state.
- * An observed component that depends only on the current hidden state in a possibly lossy way.

What is a hidden Markov Model?



What is a hidden Markov Model?

A (Mealy-type) HMM consists of:

- * A set S $\overset{\text{"think"}}{\rightsquigarrow}$ collection of hidden states
- * A set $\mathcal{X}$ $\overset{\text{"think"}}{\rightsquigarrow}$ vocabulary of observations
- * A collection of transition matrices $(T^{(x)})_{x \in \mathcal{X}}$

$\overset{\text{"think"}}{\rightsquigarrow}$ dynamic that determines
emissions & hidden
state transitions

$$\text{where } T_{ij}^{(x)} := P(X = x, S_j = s_j \mid S_i = s_i) = P(x, s_j \mid s_i)$$

It is convenient to use the following graphical notation to visualise process transitions:



Fix an initial distribution over hidden states

$$\eta_i^{(\emptyset)} := (P(s_i))_i, \quad i = 1, \dots, |S|$$

The probability of observing the sequence of emissions $w := w_1 w_2 \cdots w_n \in \mathcal{X}^n$ factorises as:

$$P(w) = \sum_{s_0, \dots, s_n} P(s_0) P(w_1, s_1 | s_0) P(w_2, s_2 | s_1) \cdots P(w_n, s_n | s_{n-1})$$

Conveniently, the above expression can be stated in terms of HMM transition matrices as:

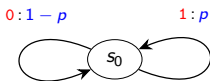
$$P(w) = \eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} \mathbb{1}$$

$$\text{where } \eta^{(\emptyset)} \text{ — row vector w.r.t ordered basis of } S \quad \& \quad \mathbb{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Example 1: (Biased coin)

$$S = \{s_0\}$$

$$\mathcal{X} = \{0, 1\}$$



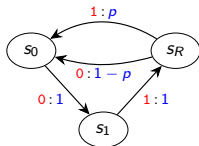
$$T^{(0)} = 1 - p$$

$$T^{(1)} = p$$

Example 2: (zero-one-random)

$$S = \{s_0, s_1, s_R\}$$

$$\mathcal{X} = \{0, 1\}$$



$$T^{(0)} = \begin{array}{c} \begin{array}{ccc|c} & s_0 & s_1 & s_R & \\ \hline s_0 & 0 & 1 & 0 & s_0 \\ s_1 & 0 & 0 & 0 & s_1 \\ s_R & 1-p & 0 & 0 & s_R \end{array} \end{array}$$

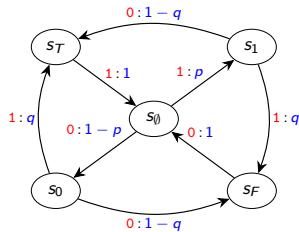
$$T^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ p & 0 & 0 \end{bmatrix}$$

$$\dots 01R01\dots \quad R \in \{0, 1\}$$

Example 3: (random-random-XOR)

$$S = \{s_\emptyset, s_0, s_1, s_F, s_T\}$$

$$\mathcal{X} = \{0, 1\}$$



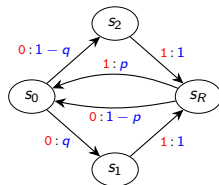
$$T^{(0)} = \begin{array}{c|cc|cc|c} & s_\emptyset & s_0 & s_1 & s_F & s_T & \\ \hline s_\emptyset & 0 & 1-p & 0 & 0 & 0 & \\ \hline s_0 & 0 & 0 & 0 & 1-q & 0 & \\ \hline s_1 & 0 & 0 & 0 & 0 & 1-q & \\ \hline s_F & 1 & 0 & 0 & 0 & 0 & \\ \hline s_T & 0 & 0 & 0 & 0 & 0 & \end{array}$$

$$T^{(1)} = \begin{array}{c|cc|cc|c} & 0 & 0 & p & 0 & 0 & \\ \hline & 0 & 0 & 0 & 0 & q & \\ \hline & 0 & 0 & 0 & q & 0 & \\ \hline & 0 & 0 & 0 & 0 & 0 & \\ \hline & 1 & 0 & 0 & 0 & 0 & \end{array}$$

Example 4:

$$S = \{s_0, s_1, s_2, s_R\}$$

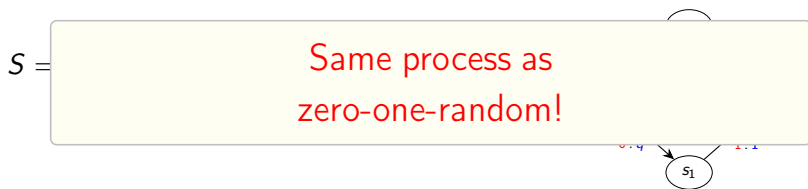
$$\mathcal{X} = \{0, 1\}$$



$$\tilde{T}^{(0)} = \begin{array}{c|cccc} & s_0 & s_1 & s_2 & s_R \\ \hline s_0 & 0 & q & 1-q & 0 \\ s_1 & 0 & 0 & 0 & 0 \\ s_2 & 0 & 0 & 0 & 0 \\ s_R & 1-p & 0 & 0 & 0 \end{array}$$

$$\tilde{T}^{(1)} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ p & 0 & 0 & 0 \end{bmatrix}$$

Example 4:



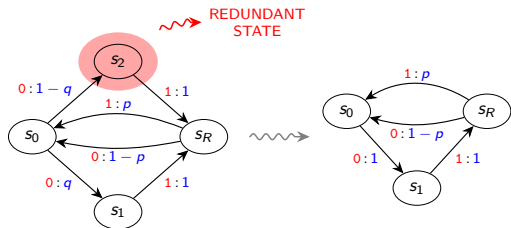
$$\tilde{T}^{(0)} = \begin{array}{c|cccc} & s_0 & s_1 & s_2 & s_R \\ \hline s_0 & 0 & q & 1-q & 0 \\ s_1 & 0 & 0 & 0 & 0 \\ s_2 & 0 & 0 & 0 & 0 \\ s_R & 1-p & 0 & 0 & 0 \end{array}$$

$$\tilde{T}^{(1)} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ p & 0 & 0 & 0 \end{bmatrix}$$

Example 4:

$$S = \{s_0, s_1, s_2, s_R\}$$

$$\mathcal{X} = \{0, 1\}$$



$$\tilde{T}^{(0)} = \begin{array}{c|cccc} & s_0 & s_1 & s_2 & s_R \\ \hline s_0 & 0 & q & 1-q & 0 \\ s_1 & 0 & 0 & 0 & 0 \\ s_2 & 0 & 0 & 0 & 0 \\ s_R & 1-p & 0 & 0 & 0 \end{array}$$

$$\tilde{T}^{(1)} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ p & 0 & 0 & 0 \end{bmatrix}$$

Let $(T^{(x)})_{x \in \mathcal{X}} - \text{Z1R}$

$(\tilde{T}^{(x)})_{x \in \mathcal{X}} - \text{Z1R with redundancy}$

There exist matrices U & V such that:

$$\tilde{T}^{(x)} = V T^{(x)} U \quad \& \quad \text{Id}_{3 \times 3} = UV$$

$$\eta^{(\emptyset)} U - \text{stochastic} \quad \& \quad V \mathbb{1} - \text{all ones}$$

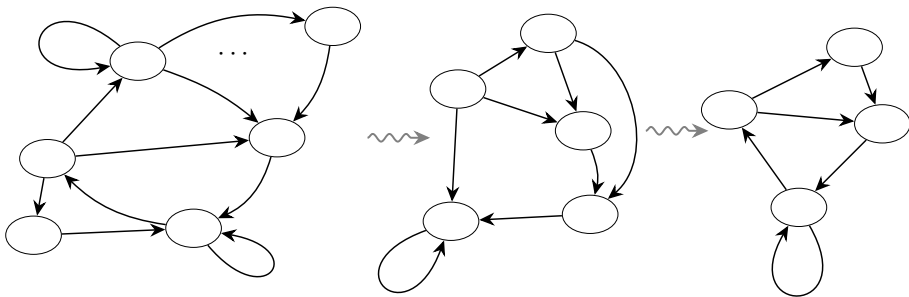
for all $x \in \{0, 1\}$. It follows that

$$\begin{aligned} & \eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \mathbb{1} \\ = & \eta^{(\emptyset)} U V T^{(w_1)} U V T^{(w_2)} U \dots V T^{(w_n)} U V \mathbb{1} \\ = & \tilde{\eta}^{(\emptyset)} \tilde{T}^{(w_1)} \tilde{T}^{(w_2)} \dots \tilde{T}^{(w_n)} \tilde{\mathbb{1}} \end{aligned}$$

HMM minimality & uniqueness

A given HMM realisation of a stochastic process may be needlessly large i.e. have more hidden states than necessary.

We say that a HMM realisation of a stochastic process is minimal if there exists no other HMM realisations with fewer hidden states.



Claim: For a stochastic process that admits an HMM realisation, the minimal HMM is not unique.

(I.e. there does not exist similarity transforms relating the minimal HMMs.)

This is quite dissatisfying!

Suppose we trained a transformer on a process whose minimal HMM is not unique; what realisation would it learn the details of?

Can we identify a simple generalisation of HMMs such that uniqueness is guaranteed for the more general class?

Consider the output distribution of an HMM with $(T^{(x)})_{x \in \mathcal{X}}$ & initial hidden state dist. $\eta^{(\emptyset)}$

$$\begin{aligned}
 P(w) &= \eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \mathbb{1} \\
 &= \eta^{(\emptyset)} V^{-1} V T^{(w_1)} V^{-1} V T^{(w_2)} V^{-1} \dots V T^{(w_n)} V^{-1} V \mathbb{1} \\
 &= \check{\eta}^{(\emptyset)} \check{T}^{(w_1)} \check{T}^{(w_2)} \dots \check{T}^{(w_n)} \check{\mathbb{1}}, \quad V \in GL_{|S|}
 \end{aligned}$$

where $\check{T}^{(x)} = V T^{(x)} V^{-1}$ & $\check{\eta}^{(\emptyset)} = \eta^{(\emptyset)} V^{-1}$ & $\check{\mathbb{1}} = V \mathbb{1}$

An identical process is produced by the transition matrices $(\check{T}^{(x)})_{x \in \mathcal{X}}$ and the vectors $\check{\eta}^{(\emptyset)}$ & $\check{\mathbb{1}}$.

$\implies$ Why constrain the transition matrix to have probabilities as matrix elements?

Generalised hidden Markov Model

A generalised hidden Markov Model (GHMM) consists of:

- * A set $\mathcal{X}$
- * A collection of transition matrices $(T^{(x)})_{x \in \mathcal{X}} \rightsquigarrow$ not necessarily sub-stochastic!
- * An initial row vector $\eta^{(\emptyset)} \rightsquigarrow$ not necessarily stochastic!

such that:

1. There exists a (column) vector ϕ satisfying:

a) $T\phi = \phi$ where $T := \sum_{x \in \mathcal{X}} T^{(x)}$

b) $\eta^{(\emptyset)}\phi = 1$

2. For all $w \in \mathcal{X}^*$, $\eta^{(\emptyset)} T^{(w_1)} \dots T^{(w_n)} \phi \geq 0$.

The thrust of this definition is to afford additional freedom to $(T^{(x)})_{x \in \mathcal{X}}$ and $\eta^{(\emptyset)}$ while maintaining the same expression assigning probabilities to emission sequences:

$$P(w) = \eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \phi$$

(This fact will be explored in the exercises.)

A consequence of this freedom is that we lose connection to explicit hidden states!

Example: (Bloch Walk)

Fix $\gamma = (2\sqrt{\alpha^2 + \beta^2})^{-1}$, where $\alpha > 0$ & $\beta \in \mathbb{R}$

$$T^{(0)} = \begin{bmatrix} \frac{1}{4} & 0 & 2\alpha\beta\gamma^2 \\ 0 & (\alpha^2 - \beta^2)\gamma^2 & 0 \\ 2\alpha\beta\gamma^2 & 0 & \frac{1}{4} \end{bmatrix}$$

$$T^{(1)} = \begin{bmatrix} \frac{1}{4} & 0 & -2\alpha\beta\gamma^2 \\ 0 & (\alpha^2 - \beta^2)\gamma^2 & 0 \\ -2\alpha\beta\gamma^2 & 0 & \frac{1}{4} \end{bmatrix}$$

$$T^{(2)} = \begin{bmatrix} \frac{1}{4} & 2\alpha\beta\gamma^2 & 0 \\ 2\alpha\beta\gamma^2 & \frac{1}{4} & 0 \\ 0 & 0 & (\alpha^2 - \beta^2)\gamma^2 \end{bmatrix}$$

$$T^{(3)} = \begin{bmatrix} \frac{1}{4} & -2\alpha\beta\gamma^2 & 0 \\ -2\alpha\beta\gamma^2 & \frac{1}{4} & 0 \\ 0 & 0 & (\alpha^2 - \beta^2)\gamma^2 \end{bmatrix}$$

$$\eta^{(\emptyset)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\phi = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Returning to our initial motivation...

Claim (Upper '97): For a stochastic process that admits a GHMM realisation, the minimal GHMM is unique (up to similarity transformation).

⇒ Transformers trained on data emitted from a GHMM should learn the structure corresponding to the minimal GHMM

We will discuss this 'structure' in the next block!

Questions?